CSA0609-DESIGN AND ANALYSIS OF ALGORITHMS

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Exp-12: Maximum and Minimum Using Divide and Conquer

Aim:

To find the maximum and minimum elements in an array using the Divide and Conquer technique.

Algorithm:

1. Read the array elements.
2. Divide the array into two halves.
3. Recursively find the maximum and minimum in each half.
4. Compare the two maximum values to find the overall maximum.
5. Compare the two minimum values to find the overall minimum.
6. Display the maximum and minimum values.

Source Code:

```
def max_min(arr, low, high):
    if low == high:
        return arr[low], arr[low]
    if high == low + 1:
        if arr[low] > arr[high]:
            return arr[low], arr[high]
        else:
            return arr[high], arr[low]
    mid = (low + high) // 2
    max1, min1 = max_min(arr, low, mid)
    max2, min2 = max_min(arr, mid + 1, high)
    maximum = max(max1, max2)
```

```
    minimum = min(min1, min2)
    return maximum, minimum
n = int(input("Enter number of elements: "))
arr = list(map(int, input("Enter array elements: ").split()))
maximum, minimum = max_min(arr, 0, n - 1)
print("Maximum:", maximum)
print("Minimum:", minimum)
print("Time Complexity: O(n)")
print("Space Complexity: O(log n)")
```

Output:

```
Enter number of elements: 6
Enter array elements: 25 10 45 5 30 15
Maximum: 45
Minimum: 5
Time Complexity: O(n)
Space Complexity: O(log n)
```

Result:

The maximum and minimum elements of the array were successfully found using Divide and Conquer.